

Bluetooth Channel Sounding

Campaign Configuration and Scheduling

Capability negotiation, configuration and procedure parameters, ACL anchoring, and the procedure–event–subevent–step hierarchy

The three companion reports treat the single CS step: what Mode-0, Mode-1 and Mode-2 measure and how the measurement is estimated. This report treats everything that surrounds the step. It follows the configuration lifecycle from capability exchange through security establishment, configuration creation, channel classification and procedure parameters to procedure enable, identifying at each stage which parameters are negotiated with the peer over the ACL connection, which are purely local, and which are *answered* by the Controller rather than requested by the Host. It then shows how the ACL connection enters the schedule: as the time base whose anchor points every CS event hangs from, as the unit in which event and procedure intervals are counted, as the source of the security context from which the CS nonce material is derived, and as the supervision relationship whose loss terminates the campaign. The scheduling hierarchy is then developed level by level – campaign, procedure, event, subevent, step – with the ordering rules that place Mode-0, main mode and sub mode steps inside a subevent, the step-duration arithmetic for all four modes, and the capacity calculation that converts a subevent length into a step count. A worked example carries one complete configuration from Host parameters to measurement rate, duty cycle and channel coverage. Coexistence, result reporting, campaign-level error sources, a verification plan and an HCI parameter appendix close the report.

Primary references	Core 6.3, Vol. 6, Part H, Secs. 4.1–4.2 and 4.5; Vol. 6, Part B, CS Link Layer control procedures
Host interface	Core 6.3, Vol. 4, Part E, LE CS command and event set
Companion reports	<i>Bluetooth CS Mode-0 Frequency Estimation</i> , Revision 3.0; <i>Bluetooth CS Mode-1 RTT Estimation</i> , Revision 1.0; <i>Bluetooth CS Mode-2 Phase-Based Ranging</i> , Revision 1.0
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Abstract

A Channel Sounding measurement campaign is the outer envelope of everything the mode reports describe. The individual step is where a frequency, a time or a phase is measured; but how many steps run, which channels they occupy, at what instants they occur, how many of them share a coherent frequency reference, and how often the whole pattern repeats are all settled before any step runs. They are settled by a configuration negotiated with the peer over the ACL connection and by a schedule that the Controller derives from that connection's own timing and reports back to the Host.

This report treats the campaign as the object of study. Three questions organise it. First, *how is a campaign configured?* The answer is a lifecycle of eight Host-initiated stages, of which four cause Link Layer control exchanges with the peer and four are local. The report tabulates every parameter of the configuration and procedure-parameter sets with its range, its units and its negotiation status. Second, *how do the ACL connection's parameters enter?* The answer is that they enter in four distinct ways: as the time base, because a CS procedure starts at a stated offset from a stated ACL connection event anchor point; as the counting unit, because the event and procedure intervals are expressed in whole connection intervals and therefore quantise the measurement rate; as the security context, because the CS nonce and personalisation material are derived from the connection's established keys and an unencrypted connection cannot carry a secured campaign; and as the liveness relationship, because peripheral latency, connection parameter updates and supervision timeout each bound or terminate the campaign. Third, *how are the steps of the different modes scheduled?* The answer is a strict intra-subevent ordering – one to three Mode-0 steps first, then main mode steps with sub mode steps inserted at a controlled cadence, with a configurable number of steps repeated from the previous subevent for continuity – combined with a per-mode step duration and a fixed inter-step frequency-change period, which together determine how many steps fit in the subevent length the Controller granted.

The report closes with a worked example that takes a 50 ms connection interval and a Mode-2 main mode with a Mode-1 sub mode from Host request to an update rate of one procedure per 400 ms at a 3.6 % radio duty cycle, a campaign-level error discussion covering staleness, target motion and cross-subevent coherence, a verification plan, and an appendix tabulating the Host interface.

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1 Scope and objective

1.1 What this report covers

A Channel Sounding measurement does not exist as a single event. It exists as a nested structure of scheduled radio activity whose innermost element – the CS step – is the only place where a physical quantity is measured, and whose outer elements exist entirely to place those steps in time, in frequency and in a shared reference frame. The companion reports analyse the innermost element. This report analyses the four levels above it and the configuration machinery that establishes them.

The report answers three questions in order.

1. **Configuration.** What must be exchanged, agreed and committed before a campaign can run, in what order, and which parameter is decided by whom? Section 3.
2. **ACL dependence.** Which parameters of the underlying ACL connection enter the CS schedule, and through which mechanism does each of them enter? Section 4.
3. **Scheduling.** How are procedures, events, subevents and steps laid out in time, and how are steps of the different modes ordered and sized inside a subevent? Sections 5 to 7.

1.2 The term “campaign”

The specification defines the CS *procedure*, the CS *event*, the CS *subevent* and the CS *step*. It does not define a term for the repeated sequence of procedures that one `Procedure.Enable` authorises and that runs until it is disabled, exhausted, or aborted. This report uses *campaign* for that outermost object, because the ranging application reasons about it directly: a campaign has a lifetime, an update rate, an energy cost and a security context, none of which is a property of any single procedure. The term is descriptive, not normative, and is used consistently in the sense of Table 2.

1.3 What the campaign determines that the step cannot

Three properties of a ranging result are decided entirely at campaign level, and no amount of per-step estimator quality recovers them if they are decided badly.

Frequency aperture. The Mode-2 report shows that ranging precision from a phase-slope fit scales with the span and count of the channels measured. Those channels come from the channel map and the channel-selection sequence, and how many of them a procedure reaches before it ends is a scheduling outcome, not an estimator outcome. A configuration that covers 20 channels per procedure and one that covers 72 differ in ranging standard deviation by a factor that no estimator choice closes.

Coherence. Mode-0 establishes a frequency reference *per subevent*. Every step in a subevent shares that reference; steps in different subevents do not. Whether a procedure’s measurements can be combined coherently therefore depends on whether they fall in one subevent or many – a scheduling decision made by the Controller when it converts the Host’s requested subevent length into a granted one.

Staleness. A distance estimate is attributed to an instant. A procedure that spans 30 ms against a target moving at 1.5 m/s smears 4.5 cm of true displacement across the measurement aperture. The procedure length is a scheduling parameter.

The design boundary. The Host requests *ranges*; the Controller returns a *schedule*. Almost every scheduling parameter in Channel Sounding is expressed as a minimum and a maximum in the Host's request and as a single settled value in the Controller's reply. A Host that treats its requested values as the operating values will mis-predict its own measurement rate, its own duty cycle and its own coherence structure. The settled values reported at procedure enable are the only ones that describe what will actually happen on air.

2 Normative framework and terminology

Table 1: Where each part of the campaign is defined. Section numbers follow the revision cited on the title page; read the clause structure from the revision in force.

Source	What it defines	Question answered
Core 6.3, Vol. 1, Part A, Sec. 9.1	Architectural role of Channel Sounding and its dependence on a connection	Why is an ACL connection a precondition?
Core 6.3, Vol. 6, Part H, Sec. 4.1–4.2	CS procedure, event, subevent and step; step ordering; scheduling relationships	How is radio activity laid out in time?
Core 6.3, Vol. 6, Part H, Sec. 4.3	Per-mode step structure and the step timing parameters	How long is one step of each mode?
Core 6.3, Vol. 6, Part H, Sec. 4.5	Antenna configuration, path permutation, channel selection	Which channel and which antenna path does a step use?
Core 6.3, Vol. 6, Part B	CS Link Layer control procedures and their PDUs	What is negotiated with the peer, and when?
Core 6.3, Vol. 4, Part E	LE CS HCI commands and events	What does the Host set, and what is reported back?
Core 6.3, Vol. 6, Part A, Sec. 3.5	Frequency generation for CS transmissions	How is the configured channel realised on air?

2.1 The five levels

Table 2 names the five levels used throughout this report and states, for each, the parameter that fixes it. The table should be read downwards: each level is a subdivision of the one above, and each is fixed by a different part of the configuration.

2.2 Notation

Table 3 collects the campaign-level symbols. Units are given because unit confusion between 0.625 ms, 1.25 ms, 625 μ s and whole connection intervals is the most frequent arithmetic error in scheduling code; Appendix A repeats them as a single reference.

2.3 Roles and the asymmetry of the campaign

The initiator and the reflector are CS roles, and they are independent of the ACL Central and Peripheral roles: either LL role may take either CS role, subject to what each device declares it supports. The asymmetry that matters for scheduling is narrower than it first appears.

Table 2: The campaign hierarchy. Only the step performs a measurement; every level above it exists to place steps in time and to bound the reference frame they share.

Level	Definition	Fixed by
Campaign	The sequence of CS procedures authorised by one enable, on one configuration, on one connection	Procedure_Count , and the enable/disable commands
CS procedure	One complete traversal of the configured channel sequence; the unit to which a distance estimate is normally attributed	Max_Procedure_Len , channel map and its repetition
CS event	The activity anchored to one ACL connection event anchor point; contains one or more subevents	Event_Interval , in whole connection intervals
CS subevent	A contiguous block of steps sharing one Mode-0 frequency reference	Subevent_Len , Subevent_Interval , Subevents_Per_Event
CS step	One reciprocal exchange on one channel in one mode; the measurement itself	Mode, step timing parameters, T_{FCS}

Both devices must hold a configuration with the same content, and both must be enabled. But the *schedule* is proposed by one side. The initiator's Controller is the side that selects the concrete anchor, offsets and intervals within the ranges its Host requested and within what the peer signals it can accept; the reflector's Controller either accepts that proposal or declines the procedure. Consequently a reflector Host that sets aggressive procedure parameters is expressing a constraint, not a schedule, and it will discover the actual schedule only in the reply to its own enable command. Section 4.5 develops this.

Table 3: Symbols used for campaign-level quantities. Where the specification names an HCI field, the field name is given; the symbol is this report’s shorthand.

Symbol	HCI field	Meaning	Units
T_C	(ACL) Connection_Interval	ACL connection interval	1.25 ms
T_{SE}	Subevent_Len	Granted subevent length	μs
T_{SEI}	Subevent_Interval	Start-to-start spacing of subevents in an event	625 μs
N_{SE}	Subevents_Per_Event	Subevents in one CS event	count
N_{EI}	Event_Interval	Spacing of CS events	T_C
N_{PI}	Procedure_Interval	Spacing of CS procedures	T_C
N_{PC}	Procedure_Count	Procedures in the campaign	count
T_{PL}	Max_Procedure_Len	Granted upper bound on procedure duration	0.625 ms
N_0	Mode_0_Steps	Mode-0 steps at the head of each subevent	count
R_m	Main_Mode_Repetition	Main mode steps repeated from the previous subevent	count
$n_{\min}, n_{\max}$	Min/Max_Main_Mode_Steps	Main mode steps between sub mode steps	count
N_{step}	(reported per subevent)	Steps actually executed in a subevent	count
$T_{step,m}$	–	Duration of one step of mode m	μs
T_{FCS}	–	Frequency-change period between steps	μs
N_{AP}	from Tone_Antenna_ Config_Selection	Antenna paths	count

3 The configuration lifecycle

3.1 The eight stages

Figure 1 shows the full path from an established connection to a running campaign. Four stages generate Link Layer control exchanges with the peer and therefore consume ACL bandwidth and are subject to the connection’s own reliability; four are local to the device and complete immediately. The distinction matters for start-up latency: a cold start that must perform capability exchange, security establishment, FAE table retrieval and configuration creation costs several connection intervals of round trips before the first step can be scheduled.

3.2 Stage 0: the ACL connection as precondition

Channel Sounding has no connectionless form. Two devices that are not connected cannot range, and the reason is not merely administrative: every mechanism the campaign depends on is a property of the connection. The connection supplies a common time base with a known anchor, a reliable bidirectional control channel for negotiating the configuration, an established security context, an agreed PHY and data rate for that control channel, and a supervision relationship that detects loss. The connection also supplies the identity binding: a CS result is meaningful only if it is known which peer it describes, and the connection handle is that binding.

Encryption is the practical precondition for a secured campaign. Security establishment

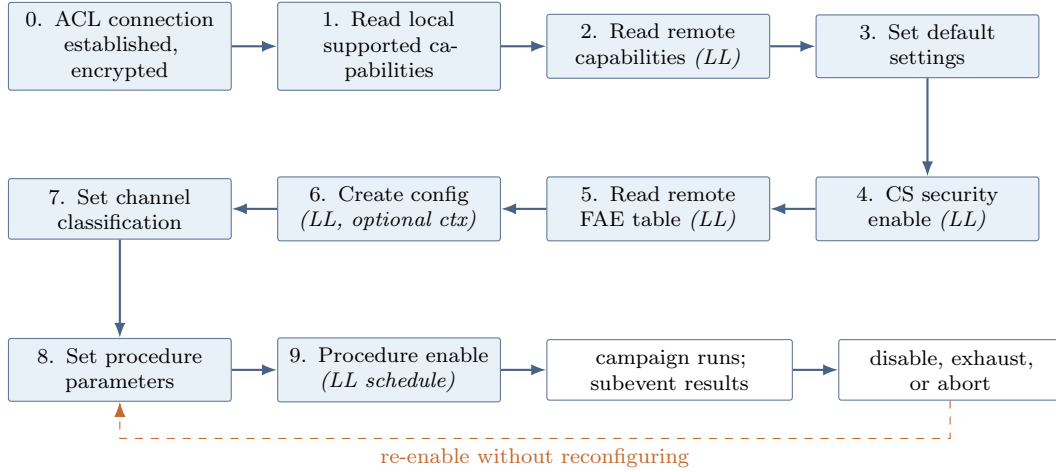


Figure 1: Configuration lifecycle. Italicised stages generate Link Layer control exchanges with the peer. Stages 8 and 9 may be repeated for the lifetime of the configuration; stages 1–7 need not be repeated unless capabilities, security context or configuration content change.

(Section 3.5) derives CS-specific material from the connection’s established keys; on an unencrypted connection it cannot proceed, and the campaign that results – if the implementation permits one at all – carries randomised sequences that are not cryptographically bound to the peer, which is precisely the condition the sounding sequence and the attack detector exist to prevent.

3.3 Stages 1–2: capability exchange

The local Controller reports what it can do; the remote Controller’s capabilities are then fetched across the connection with a Link Layer exchange and cached. Every subsequent parameter choice is constrained by the intersection of the two sets. Table 4 lists the capability fields that bear on configuration and scheduling.

Capability exchange is a scheduling input, not a formality. T_{SY} follows from CS_SYNC_PHY; N_{AP} follows from the antenna capability of *both* sides; T_{IP1} , T_{IP2} , T_{SW} , T_{PM} and T_{FCS} follow from the intersection of the two supported-value bitmaps. Every one of those appears in the step-duration equations of Section 7. A peer that supports only $T_{FCS} \geq 100 \mu s$ has already reduced the number of steps that will fit in any granted subevent, before a single scheduling parameter has been chosen.

3.4 Stage 3: default settings

The default-settings stage is local and cheap, and it carries three things: which CS roles this device will accept for this connection, which antenna the device will use for the CS_SYNC packets, and the maximum transmit power the Controller may select for CS transmissions. The last is not merely a regulatory or power-budget control: it interacts with the reported per-step signal quality and, through it, with the weighting used when steps are aggregated. A campaign run at a power ceiling below the link’s needs produces steps that pass their validity gates less often, which reduces the effective aperture of every procedure.

3.5 Stage 4: security establishment

The CS security start procedure derives, from the connection’s established security context, the CS-specific material that personalises the campaign: an initialisation vector, an instan-

Table 4: Capability fields that constrain the campaign. Field names follow the Host interface; the specification’s table is authoritative for ranges and for fields omitted here.

Capability	Representative range	What it constrains
Num_Config_Supported	1–4	How many configurations may coexist on one connection
Max_Consecutive_Procedures_Supported	implementation	Upper bound on Procedure_Count
Num_Antennas_Supported	1–4	Antenna configuration indices that are legal
Max_Antenna_Paths_Supported	1–4	N_{AP} , and hence Mode-2/3 step duration
Roles_Supported	initiator, reflector	Which CS role this device may take
Optional_Modes_Supported	Mode-3	Whether Mode-3 may be a main or sub mode
RTT_Capability, RTT*_N	precision class and step counts	Which RTT types are usable and how many steps a precision claim needs
Optional_CS_SYNC_PHYs_Supported	LE 2M, LE 2M 2BT	CS_SYNC_PHY and hence T_{SY}
Optional_Subfeatures_Supported	companion signal, zero FAE, CSA #3c, PBR from RTT sequence	Channel selection type; FAE handling
T_IP1/T_IP2/T_FCS/T_PM.Times_Supported	bitmaps over the permitted value sets	The step timing parameters, and hence step duration
T_SW.Time_Supported	0, 1, 2, 4, 10 μ s	Antenna switch period in Mode-2/3
TX_SNR.Capability	bitmap	Whether SNR control may be requested

tiation nonce and a personalisation vector, conventionally denoted **CS_IV**, **CS_IN** and **CS_PV**. These seed a deterministic random bit generator whose output is consumed by the campaign in four places.

1. The 32-bit access address carried by each **CS_SYNC** packet, which differs per step and is unpredictable to a third party.
2. The sounding sequence or random sequence payload used by the Mode-1 and Mode-3 RTT types, which is what makes an early-detect/late-commit attack require prediction rather than observation.
3. The channel sequence produced by the channel selection algorithm, so that the next channel is not predictable from the previous ones.
4. The antenna path permutation applied per step in Mode-2 and Mode-3.

Two consequences follow for the campaign. First, security establishment must complete before a configuration that uses a sounding or random RTT type can be enabled; the alternative is a campaign whose sequences are not bound to the peer. Second, the derived material is a property of the connection’s security context: a re-pairing, a key refresh or a reconnection invalidates it, and the campaign must be re-established. An implementation that caches a configuration across a reconnection must not cache the security material with it.

3.6 Stage 5: the FAE table

The remote device’s frequency actuation error table gives, per CS channel, the known systematic offset between the frequency the peer intends to transmit and the frequency it actually transmits. The Mode-0 report treats its use. At campaign level the relevant facts are that the table is fetched once over the connection and cached, that a device declaring the zero-frequency-actuation-error subfeature obviates the fetch, and that a stale cached table is a silent bias: it is not re-validated during a campaign, so a peer whose calibration changes with temperature over a long campaign will not be corrected until the table is fetched again.

3.7 Stage 6: creating a configuration

The configuration is the content of the measurement: which modes, which RTT type, which sync PHY, which channels, how the steps are ordered. It is not the schedule. Table 5 lists the parameter set.

Table 5: Configuration parameters. “Negotiated” means the value is carried to the peer in a Link Layer configuration exchange and must be agreed; “local” means it applies to this device only. Ranges are representative of the cited revision.

Parameter	Range	Status	Effect
Config_ID	0–3	local	Selects one of the coexisting configurations
Create_Context	local only / local and remote	local	Whether the peer is configured by this command
Main_Mode_Type	1, 2, 3	negotiated	The mode most steps use
Sub_Mode_Type	1, 2, 3, none	negotiated	The interleaved secondary mode
Min_Main_Mode_Steps	2–255	negotiated	Lower bound on main mode steps between sub mode steps
Max_Main_Mode_Steps	2–255	negotiated	Upper bound on the same
Main_Mode_Repetition	0–3	negotiated	Main mode steps repeated from the previous subevent
Mode_0_Steps	1–3	negotiated	Mode-0 steps at the head of every subevent
Role	initiator / reflector	local	This device’s CS role
RTT_Type	AA only; 32/96-bit sounding; 32/64/96/128-bit random	negotiated	T_{SY} , timing precision, attack resistance
CS_SYNC_PHY	LE 1M, LE 2M, LE 2M 2BT	negotiated	T_{SY} and receiver sensitivity
Channel_Map	10 octets, ≥ 15 channels enabled	negotiated	The frequency aperture
Channel_Map_Repetition	1–255	negotiated	How many times the map is traversed per procedure
Channel_Selection_Type	#3b, #3c	negotiated	Sequence generation
Ch3c_Shape, Ch3c_Jump	hat / X; 2–8	negotiated	Shape parameters of CSA #3c

3.7.1 Legal main/sub mode combinations

Not every pair is permitted. The rule follows from what a sub mode is for: it supplies the complementary observable to the main mode, so that a procedure yields both an unambiguous coarse range and a precise ambiguous one.

Table 6: Permitted main and sub mode combinations, and what each yields.

Main mode	Sub mode	Observables per procedure	Typical intent
Mode-1	none	RTT only	Coarse ranging; presence and integrity
Mode-1	Mode-2	RTT dense, PBR sparse	RTT-led, PBR used to refine
Mode-2	none	PBR only	Precise but ambiguous; needs external disambiguation
Mode-2	Mode-1	PBR dense, RTT sparse	The common high-accuracy configuration
Mode-2	Mode-3	PBR dense, combined sparse	PBR-led with combined anchors
Mode-3	none	RTT and PBR every step	Maximum information per step, longest step
Mode-3	Mode-2	Combined dense, PBR sparse	Mode-3 led, cheaper interleaved steps

Mode-0 is not a main or sub mode. It is not selected by `Main_Mode_Type` or `Sub_Mode_Type`, and it is not optional. It is inserted at the head of every subevent by the scheduling rules, in the quantity given by `Mode_0_Steps`. This is a structural statement about the campaign, not a per-step one: the frequency reference is a *subevent* property, and the cost of establishing it is paid once per subevent, not once per procedure.

3.8 Stage 7: channel classification

Channel classification is the mechanism by which a device tells the campaign which CS channels it believes are unusable. It is distinct from the ACL connection's own channel map, and distinct from the configuration's `Channel_Map`: the configuration states which channels the campaign may use, and classification states which of them are currently believed bad. Section 9 develops the interaction, including the important asymmetry that classification updates take effect at a procedure boundary rather than immediately.

3.9 Stage 8: procedure parameters

This is where the Host expresses the schedule it wants. Every scheduling quantity here except the antenna configuration is a bound rather than a value.

3.10 Stage 9: enable, and the schedule that comes back

Enabling the procedure is the point at which the abstraction inverts. Up to here the Host has been supplying constraints; from here the Controller supplies facts. The completion event reports the settled schedule, and it is the authoritative description of what will happen on air.

Table 7: Procedure parameters. Note that six of the thirteen are minimum/maximum pairs: they express an acceptable range, not a request for a specific schedule.

Parameter	Range and units	Kind	Effect
Max.Procedure_Len	1–65535 $\times$ 0.625 ms (to 40.96 s)	bound	Upper limit on one procedure’s duration
Min.Procedure_Interval	1–65535 connection intervals	bound	Fastest acceptable repetition
Max.Procedure_Interval	1–65535 connection intervals	bound	Slowest acceptable repetition
Max.Procedure_Count	1–65535; 0 for continuous	bound	Campaign length
Min.Subevent_Len	1250–3 999 999 μ s	bound	Smallest acceptable coherent block
Max.Subevent_Len	1250–3 999 999 μ s	bound	Largest acceptable coherent block
Tone_Antenna_Config_Selection	0–7	value	N_{AP} , and Mode-2/3 step duration
PHY	LE 1M / 2M / Coded	value	PHY used for the procedure’s LL signalling
Tx.Power_Delta	dB, or “no recommendation”	value	Requested deviation from the link’s power
Preferred_Peer_Antenna	bitmap over 4 elements	value	Which peer elements to use
SNR.Control_Initiator	18/21/24/27/30 dB or off	value	Transmit power control target at the initiator
SNR.Control_Reflector	as above	value	The same at the reflector

Read the reply, then compute. Measurement rate, radio duty cycle, energy per fix, coherence structure and channel coverage per procedure are all functions of the fields in Table 8 and of the per-mode step durations – not of the ranges requested in Table 7. Section 8 performs that computation end to end for a representative configuration.

Table 8: The settled schedule reported at procedure enable. Every field is an answer, not a request.

Reported field	Units	Interpretation
Selected_TX_Power	dBm	The power the Controller will actually use
Subevent_Len	μs	Granted subevent length; the coherence block
Subevents_Per_Event	count (to 16)	Subevents packed into one CS event
Subevent_Interval	625 μs ; 0 if one subevent	Start-to-start spacing within an event
Event_Interval	connection intervals	Spacing of CS event anchors
Procedure_Interval	connection intervals	Spacing of procedure anchors
Procedure_Count	count; 0 for continuous	Campaign length as settled
Max_Procedure_Len	0.625 ms	The granted procedure length bound

4 The ACL connection as substrate

The ACL connection is not merely a prerequisite that can be satisfied and then forgotten. It enters the campaign continuously and in four distinct capacities. This section takes them in turn and closes with a summary table mapping each ACL parameter to the CS quantity it controls.

4.1 Capacity one: the time base

A CS procedure starts at a stated offset from the anchor point of a stated ACL connection event. This single sentence carries most of the coupling.

The ACL connection defines a periodic sequence of anchor points separated by the connection interval T_C , numbered by a connection event counter that both devices maintain and that survives peripheral latency and missed events. Because both Controllers already agree on that sequence to within the accuracy required to maintain the link, it is the natural common clock, and Channel Sounding uses it as such: the campaign is expressed as (*connection event number, offset*) pairs rather than as absolute times. No new synchronisation mechanism is introduced at campaign level; the sub-microsecond alignment that ranging needs is established inside each subevent by the Mode-0 steps, on top of the coarse alignment the connection already provides.

Three consequences deserve to be stated explicitly.

The connection interval is the scheduling quantum. The spacing between CS events and between CS procedures is expressed in whole connection intervals. A campaign on a 7.5 ms connection can be repeated at 7.5 ms granularity; the same campaign on a 100 ms connection can only be repeated at 100 ms granularity. A Host that wants a 250 ms update rate on a 100 ms connection cannot have it: it can have 200 ms or 300 ms.

The connection interval bounds the subevent. CS activity must coexist with the connection events themselves and with whatever else the Controller has scheduled. A subevent whose length approaches the connection interval leaves no room, and the Controller will grant less than was asked for – which, as Section 5.3 shows, silently changes the coherence structure of the measurement by splitting what the Host expected to be one subevent into several.

Anchor drift is common-mode. Because both devices track the same connection event

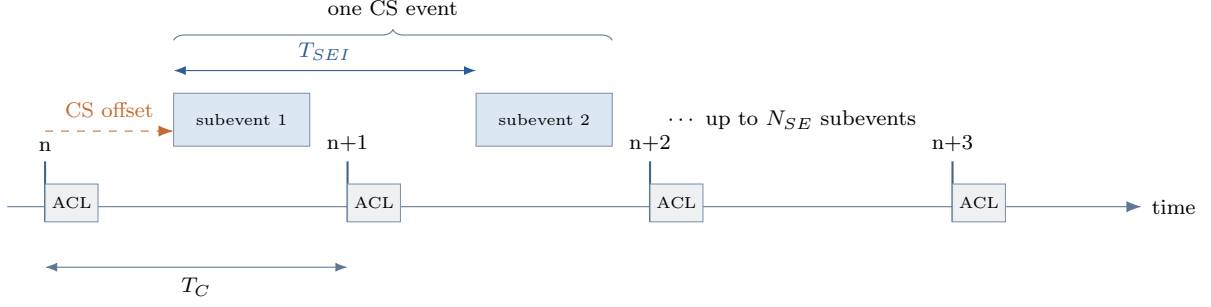


Figure 2: Anchoring. The CS event begins at a fixed offset from the ACL connection event anchor point of a nominated connection event, and its subevents are spaced by T_{SEI} . The ACL connection event itself continues to occur; CS activity is placed around it, not instead of it.

counter and the same anchor sequence, slow drift of the pair’s shared timing relative to absolute time does not disturb the campaign. What would disturb it is a change in the anchor sequence itself, which is the subject of Section 4.4.

4.2 Capacity two: the counting unit and interval arithmetic

Given the settled schedule, the campaign’s timing is fully determined:

$$\text{CS event period} = N_{EI} T_C, \quad (1)$$

$$\text{CS procedure period} = N_{PI} T_C, \quad (2)$$

$$\text{subevent starts within an event} = \{0, T_{SEI}, 2T_{SEI}, \dots, (N_{SE} - 1)T_{SEI}\}, \quad (3)$$

$$\text{campaign duration} = N_{PC} N_{PI} T_C \quad (N_{PC} \neq 0). \quad (4)$$

Two structural constraints link these. First, a procedure must be able to contain the events it needs: the number of CS events in a procedure is bounded by $\lceil N_{PI}/N_{EI} \rceil$, and if the procedure’s step budget requires more events than that, the procedure is truncated rather than extended. Second, the granted T_{PL} bounds the procedure independently of the event structure; a procedure ends when it has executed its steps, when it reaches T_{PL} , or when it is aborted, whichever comes first.

$N_{EI} = 1$ is not always available. An implementation may be unable to place CS events on every connection event – because the CS event plus the connection event plus scheduling margin does not fit in T_C , or because other links share the radio. Controllers commonly return $N_{EI} = 2$ where the Host asked for the fastest possible schedule. This is legitimate: the Host asked for a range, and the Controller answered within it. The observable effect is that the measurement rate is half what a naive calculation predicted, and the only reliable way to know is to read N_{EI} back from the enable completion.

4.3 Capacity three: the security context

Section 3.5 described the derivation. The campaign-level point is the coupling in the other direction: anything that disturbs the connection’s security context disturbs the campaign. Re-pairing, an encryption pause and restart, or a key refresh invalidate the CS material, and a correct implementation re-runs security establishment rather than continuing with stale nonce material. Continuing with stale material is not a subtle degradation – it means the access addresses and sequences of the new campaign are predictable from observation of the old one, which is the exact capability a distance-reduction attacker needs.

4.4 Capacity four: liveness, latency and updates

4.4.1 Peripheral latency

Peripheral latency permits the Peripheral to skip connection events. The anchor sequence is unaffected – the counter still advances – so the arithmetic of Equations (1)–(4) still holds. What changes is the practical one: a Peripheral that is skipping connection events to save power must nonetheless wake for the connection events that anchor CS events. A campaign therefore erodes the power saving that latency was configured to provide, and a Host that measures the energy cost of ranging must attribute the extra wake-ups to the campaign, not to the link.

4.4.2 Connection parameter updates

A connection parameter update changes T_C , and therefore changes the physical meaning of every interval expressed in connection intervals. A campaign configured for a 250 ms procedure period as $N_{PI} = 5$ on a 50 ms connection becomes a 100 ms procedure period if the connection interval is changed to 20 ms, without any CS parameter changing. Two disciplines follow: an application that depends on a particular update rate must re-read its own schedule after any connection parameter update, and a Host that intends to change connection parameters during a campaign should expect the Controller to disable, re-negotiate or reject rather than silently re-time.

4.4.3 Subrating

Connection subrating multiplies the effective connection interval for data purposes while preserving the underlying anchor sequence. Where it is in use, the relationship between the connection interval the Host configured and the anchor points actually available for CS anchoring is no longer the naive one, and the settled N_{EI} must again be read rather than assumed.

4.4.4 Supervision timeout and termination

The campaign is subordinate to the connection. Loss of the connection terminates the campaign immediately and without a procedure completion; the Host observes a disconnection, not a CS event. The supervision timeout therefore sets the worst-case time for which a ranging application may continue to believe a campaign is running after the peer has in fact gone. Applications that gate physical access on proximity should treat the supervision timeout, not the procedure interval, as the staleness bound on the last result.

4.5 The scheduling negotiation itself

The Link Layer carries a small family of CS control procedures over the ACL connection. Their names follow the usual LL_CS_* pattern and their functions divide cleanly:

The third row from the bottom is the one that answers the question of who decides the schedule. The initiator's Controller composes a proposal from its Host's requested ranges, its own scheduling constraints and the peer's declared capabilities, and offers it – typically as a set of acceptable offsets rather than a single instant, so the reflector has room to fit the campaign around its own commitments. The reflector's Controller selects within what was offered and replies; the initiator confirms. Only then do both Controllers report the settled schedule to their respective Hosts.

Table 9: Link Layer control exchanges used by the campaign, and what each carries. PDU names follow the cited revision.

Exchange	Stage	Carries
LL_CS_CAPABILITIES_REQ/RSP	2	The capability set of Table 4
LL_CS_SEC_REQ/RSP	4	Material for deriving CS_IV, CS_IN, CS_PV
LL_CS_FAE_REQ/RSP	5	The per-channel frequency actuation error table
LL_CS_CONFIG_REQ/RSP	6	The configuration content of Table 5
LL_CS_CHANNEL_MAP_IND	7	Updated channel classification
LL_CS_REQ / RSP / IND	9	The concrete schedule: anchor connection event, offset, subevent length and interval, event and procedure intervals
LL_CS_TERMINATE_IND	any	Orderly termination of a running campaign

Why a reflector Host cannot dictate the rate. The reflector's procedure parameters constrain what it will accept, and a proposal outside them is refused – which manifests as a failed enable, not as a slower campaign. A reflector that wants a faster campaign must either widen its acceptable range so that a proposal can succeed, or influence the initiator through the application layer. This is a frequent source of confusion when both sides are developed independently and each assumes its own requested interval is the operating one.

4.6 Summary: ACL parameter to CS effect

Table 10: How each parameter of the ACL connection enters the campaign. The right-hand column names the mechanism, not merely the correlation.

ACL parameter	Enters the campaign as	Mechanism
Connection interval T_C	The scheduling quantum for N_{EI} and N_{PI} ; the space available for a subevent	Equations (1)–(4); subevent must fit between connection events
Connection event counter	The shared timeline on which the campaign is expressed	Procedure start is (event number, offset)
Connection event anchor point	The origin of the CS offset	Figure 2
Peripheral latency	Wake-up obligations that the campaign reinstates	CS events anchor to connection events that latency would otherwise skip
Supervision timeout	Worst-case staleness bound on the last result	Campaign ends silently with the connection
Encryption and key material	The root of CS_IV, CS_IN, CS_PV	Security establishment, Section 3.5
Connection PHY	The rate of the control exchanges, hence configuration latency	Stages 2, 4, 5, 6, 9 are LL round trips
Connection subrating	Alters which anchors are practically available	N_{EI} granted by the Controller
ACL channel map	Independent of the CS channel map, but a proxy for the same interference	Classification input, Section 9
Connection handle	Identity binding of every CS result	All CS commands and events are per handle

5 The scheduling hierarchy

5.1 The whole structure in one picture

Figure 3 shows the five levels on one time axis, with each level expanded from the one above. The figure is the reference for the rest of the report: every scheduling parameter names one of its dimensions.

5.2 The step

The step is the atom. It occupies one channel, uses one mode, and consists of a reciprocal exchange whose structure is given in the companion reports. Between consecutive scheduled steps the radio must retune, and the frequency-change period T_{FCS} is reserved for that purpose. T_{FCS} is a configuration-negotiated value drawn from a permitted set (representatively 15, 20, 30, 40, 50, 60, 80, 100, 120 and 150 μ s) and it is not measurement time: it is pure overhead that scales with the step count, and at the fast end of the range it is a small fraction of a step while at the slow end it can approach half of a Mode-1 step.

5.3 The subevent

The subevent is the coherence block, and it is the most consequential level of the hierarchy for estimation quality.

A subevent begins with $N_0 \in \{1, 2, 3\}$ Mode-0 steps, whose combined result is the

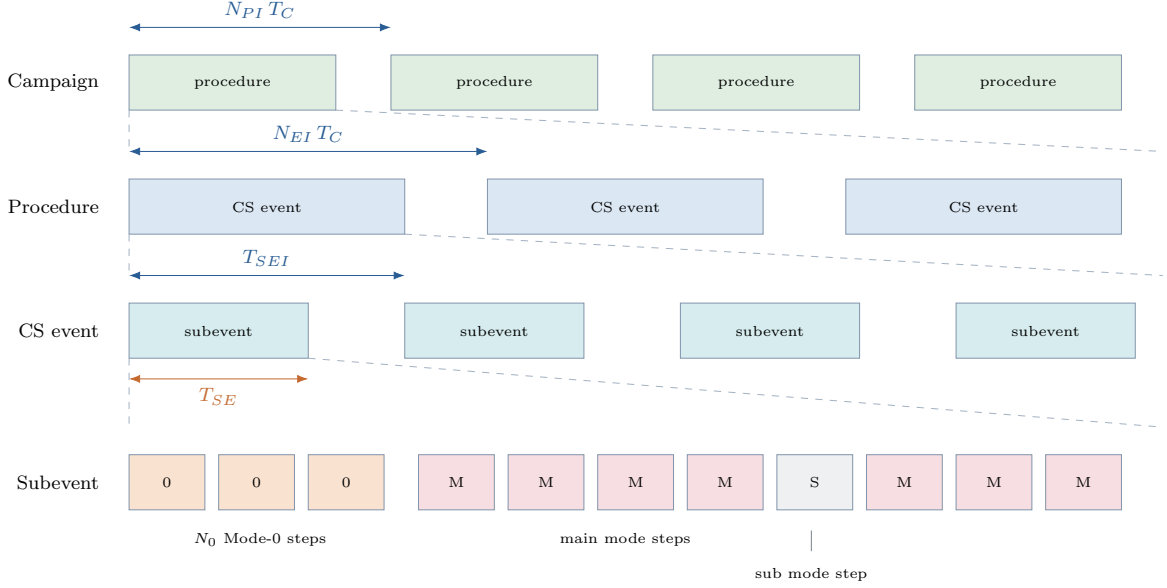


Figure 3: The campaign hierarchy on one axis. Each level is an expansion of one box of the level above. M denotes a main mode step, S a sub mode step, 0 a Mode-0 step. Widths are illustrative; the per-mode step durations of Section 7 differ substantially between modes.

subevent frequency reference FFO_{RX}^* . Every subsequent step in that subevent is processed against that reference. Steps in a different subevent are processed against a different reference, estimated at a different time from different noise. Two practical rules follow.

Phase measurements do not cross subevent boundaries for free. A Mode-2 phase-slope fit run across channels drawn from two subevents inherits the difference between the two frequency references as an additional term. That term is small when the references are good, but it is not zero, and it is not modelled by a fit that assumes one common reference. Where the highest accuracy is required, the per-subevent fits should be formed first and combined afterwards, or the configuration should be arranged so that a procedure’s PBR steps fall in a single subevent.

Requesting a long subevent is requesting coherence, not just capacity. The Host’s `Min.Subevent.Len` is the mechanism by which it says “do not split my measurement”. If the Controller cannot grant it, the enable either fails or returns a shorter subevent – and a shorter subevent means more subevents, more Mode-0 overhead, and a fragmented coherence structure.

The subevent also carries a fixed cost. The N_0 Mode-0 steps are the longest steps in the campaign (Section 7), and they measure no distance. The Mode-0 overhead fraction is

$$\eta_0 = \frac{N_0 (T_{step,0} + T_{FCS})}{T_{SE}}, \quad (5)$$

which for a short subevent can exceed a fifth of the available time. Choosing $N_0 = 3$ for robustness and then requesting a short subevent is a self-defeating combination.

5.4 The event

The event is the scheduling container that binds subevents to one ACL anchor. Its parameters are N_{SE} , the number of subevents it holds, and T_{SEI} , their spacing. Where $N_{SE} = 1$ the subevent interval is meaningless and is reported as zero.

The event exists because the useful measurement block and the connection's natural rhythm are different sizes. A single subevent of a few hundred microseconds is far shorter than a 50 ms connection interval; packing several subevents into one event lets the campaign use the interval efficiently without demanding a CS event at every connection event. Conversely, where the connection interval is short, $N_{SE} = 1$ and $N_{EI} > 1$ is the natural arrangement.

Subevent interval is quantised to 625 μ s. Like several other LE scheduling quantities, T_{SEI} is counted in 625 μ s units. A Controller cannot place subevents 400 μ s apart, so a subevent that is granted 300 μ s of length still occupies a 625 μ s cadence slot when $N_{SE} > 1$. The gap is real idle time and should be counted as such in energy models.

5.5 The procedure

The procedure is the natural unit of a ranging result. Its content is defined by the channel sequence: a procedure traverses the enabled channels of the configuration's channel map, repeated `Channel_Map_Repetition` times, in the order produced by the selected channel selection algorithm. When the sequence is exhausted the procedure ends.

Three bounds can end a procedure before its sequence is exhausted:

1. the granted T_{PL} , in 0.625 ms units, elapses;
2. the events available before the next procedure anchor, $\lceil N_{PI}/N_{EI} \rceil$, are used up;
3. an abort condition arises (Section 10.2).

A truncated procedure is not an error and is reported as such, but it changes the frequency aperture of that result, and an aggregation layer that weights procedures equally without checking their completion status will produce inconsistent precision from one fix to the next.

5.6 The campaign

Finally, `Procedure_Count` sets how many procedures the enable authorises: a finite count for a bounded burst, or the continuous setting for a campaign that runs until explicitly disabled. Three patterns are common in practice.

Table 11: Common campaign patterns and their parameter shapes.

Pattern	Intent	N_{PC}	N_{PI}
Single shot	One distance, minimum energy	1	irrelevant
Burst	Average several procedures into one fix	4–16	smallest granted
Continuous tracking	Maintain a distance estimate	continuous	set by required update rate
Duty-cycled tracking	Presence with occasional refinement	continuous	large; widen with range

6 Step composition inside a subevent

This section answers the third question of Section 1: given a configuration, in what order do steps of the different modes actually occur?

6.1 The ordering rule

The ordering is fixed by three rules applied in sequence.

1. **Mode-0 first.** Every subevent begins with exactly N_0 Mode-0 steps, where $N_0 = \text{Mode_0_Steps} \in \{1, 2, 3\}$. They are not optional, they are not distributed through the subevent, and they are repeated in every subevent of the procedure – not once per procedure.
2. **Repetition of the previous subevent’s main mode steps.** The next $R_m = \text{Main_Mode_Repetition}$ steps repeat main mode steps from the end of the preceding subevent, on the same channels. For the first subevent of a procedure there is nothing to repeat and the rule is inert. $R_m \in \{0, 1, 2, 3\}$.
3. **Main mode steps with sub mode steps interleaved.** The remainder of the subevent is filled with main mode steps, into which sub mode steps are inserted such that the number of consecutive main mode steps between two sub mode steps lies in $[n_{\min}, n_{\max}] = [\text{Min_Main_Mode_Steps}, \text{Max_Main_Mode_Steps}]$. The exact cadence within that range is the Controller’s choice, and it need not be constant. Where no sub mode is configured, the remainder is main mode steps only.

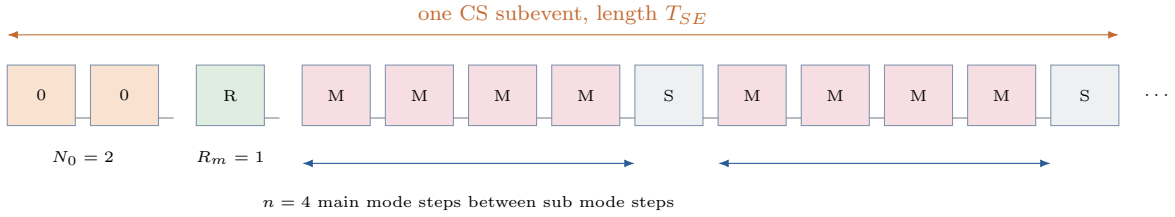


Figure 4: Step ordering inside a subevent for $N_0 = 2$, $R_m = 1$, sub mode configured, and a Controller-chosen cadence of $n = 4$. The short grey marks between steps are the frequency-change periods T_{FCS} . R denotes a repeated main mode step from the previous subevent.

6.2 What each rule is for

6.2.1 Why Mode-0 is per subevent

The Mode-0 result is a frequency reference with a finite useful life. Crystal offsets drift with temperature and supply, and the ranging chain’s tolerance for residual frequency error is tight: the Mode-2 report shows that a residual offset multiplied by the reciprocal measurement interval reads directly as distance. Re-establishing the reference at the head of each coherence block bounds the age of the reference to at most one subevent, which is the same bound that makes the subevent a valid coherence block in the first place. The choice of N_0 trades that robustness against Equation (5): $N_0 = 1$ gives one frequency estimate with no redundancy and no means of detecting a bad one; $N_0 = 3$ permits a median or a validity-gated average at a cost that Section 7 quantifies.

6.2.2 Why main mode steps are repeated across the boundary

R_m exists to create deliberate overlap. The repeated steps measure the same channels in two consecutive subevents, and therefore under two different frequency references. Their difference is a direct observation of the inter-subevent reference discontinuity discussed in

Section 5.3, and it is what allows measurements from adjacent subevents to be stitched into one aperture rather than merely averaged. $R_m = 0$ is the cheapest and is appropriate when a procedure fits in one subevent; $R_m \geq 1$ is what makes a multi-subevent procedure coherently combinable.

The cost is exact and easy to overlook: R_m steps per subevent are spent re-measuring channels already measured, so they do not extend the frequency aperture. In a procedure of S subevents the campaign spends $R_m(S - 1)$ steps on continuity.

6.2.3 Why sub mode steps are interleaved rather than blocked

A sub mode exists to supply the complementary observable, and the complementary observable is only useful if it is contemporaneous with the main one. A block of RTT steps at the end of a procedure would be separated from the early PBR steps by the whole procedure length, during which the target may have moved. Interleaving at a cadence of $n_{\min}$ to $n_{\max}$ main steps distributes the disambiguating measurement through the aperture, so that the coarse and fine observables describe the same instant to within a few steps.

The cadence is a direct trade. Table 12 gives the arithmetic.

Table 12: Sub mode cadence. The fraction of steps spent on the sub mode is $1/(n + 1)$ for a constant cadence of n main mode steps per sub mode step.

n	Sub mode fraction	Main mode fraction	Character
2	33.3 %	66.7 %	Sub mode heavily sampled; large capacity cost
4	20.0 %	80.0 %	Robust disambiguation; common choice
9	10.0 %	90.0 %	Aperture-favouring; adequate where ambiguity is small
19	5.0 %	95.0 %	Minimal disambiguation; assumes prior range knowledge

$n_{\min}$ and $n_{\max}$ are a range given to the Controller, not a schedule. A Host that needs a known number of sub mode steps per procedure should set $n_{\min} = n_{\max}$ and accept the scheduling rigidity, or – better – count the reported step modes in the subevent results rather than predicting them. The results report the mode of every step, so the realised cadence is always observable after the fact.

6.3 Channel assignment

Each step occupies one channel. The channels are consumed in the order given by the sequence that the selected channel selection algorithm generates for the procedure from the enabled channel map and the campaign’s random material, with two structural exceptions: the Mode-0 steps at the head of a subevent take their channels from the head of that subevent’s allocation, and the R_m repeated steps deliberately re-use the channels of the steps they repeat.

The consequence for aperture planning is that the number of *distinct* channels a procedure measures is not the number of steps it executes. For a procedure of S subevents:

$$N_{\text{distinct}} \leq \underbrace{S N_{\text{step}}}_{\text{steps executed}} - \underbrace{S N_0}_{\text{Mode-0}} - \underbrace{R_m(S - 1)}_{\text{repetition}}, \quad (6)$$

with equality when the channel map repetition has not caused the sequence to wrap. Section 8 evaluates this for a concrete configuration.

6.4 Modes and their scheduling character

Table 13: Scheduling character of the four step modes. Duration figures are for the representative parameter set of Section 8 and are derived in Section 7.

Mode	Role in the subevent	Typical duration	Scheduling consequence
Mode-0	Frequency and timing reference; head of every subevent	$192\ \mu\text{s}$	Fixed overhead per subevent; scales with N_0
Mode-1	RTT; main mode or sub mode	$134\ \mu\text{s}$	Cheapest step; favours long apertures
Mode-2	PBR; main mode or sub mode	$170\ \mu\text{s}$	Duration grows with N_{AP} and T_{PM}
Mode-3	RTT and PBR in one step; main mode or sub mode	$254\ \mu\text{s}$	Most information per step, fewest steps per subevent

Mode-3 deserves a specific scheduling comment. Because it carries both observables in a single step, it removes the need for a sub mode and therefore removes the interleaving question entirely: every step yields both an RTT and a per-path phase set on the same channel at the same instant, which is the ideal input to ambiguity resolution. The price is paid in capacity. At the parameters of Table 13 a Mode-3 step costs 1.9 Mode-1 steps and 1.5 Mode-2 steps, so a subevent that holds 20 Mode-2 steps holds only 14 Mode-3 steps. Whether that trade is favourable depends entirely on whether the configuration is aperture-limited or disambiguation-limited.

7 Step durations and subevent capacity

The scheduling parameters of Section 5 say how much *time* the campaign has. The step durations say how much *measurement* fits in it. This section assembles both into the capacity arithmetic.

7.1 Step duration by mode

The four step-duration expressions follow directly from the air-interface structures analysed in the companion reports. All four are reproduced here in a common notation so that they can be compared.

$$T_{step,0} = 2T_{SY}^{(0)} + 2T_{RD} + T_{IP1} + T_{GD} + T_{FM}, \quad (7)$$

$$T_{step,1} = 2T_{SY} + 2T_{RD} + T_{IP1}, \quad (8)$$

$$T_{step,2} = 2(T_{SW} + T_{PM})(N_{AP} + 1) + 2T_{RD} + T_{IP2}, \quad (9)$$

$$T_{step,3} = 2T_{SY} + 2(T_{SW} + T_{PM})(N_{AP} + 1) + 2T_{RD} + T_{IP1}. \quad (10)$$

with $T_{RD} = 5\ \mu\text{s}$, $T_{GD} = 10\ \mu\text{s}$ and $T_{FM} = 80\ \mu\text{s}$ fixed, and T_{IP1} , T_{IP2} , T_{SW} , T_{PM} and T_{FCS} configuration-negotiated within the capability intersection. Equations (7)–(9) are those of the Mode-0, Mode-1 and Mode-2 reports; Equation (10) is assembled from the combined structure and should be checked against the step-timing table of the specification revision in force before being used for a conformance claim.

Two subtleties in T_{SY} are easy to lose.

The Mode-0 packet carries no sequence. $T_{SY}^{(0)}$ is the bare preamble, access address and trailer – 44 μ son the LE 1M PHY and 26 μ son the LE 2M PHYs – regardless of the configured RTT type. The sounding or random sequence lengthens only the Mode-1 and Mode-3 packets.

T_{SY} is set by the RTT type, and the RTT type is a security choice. Moving from access-address-only to a 128-bit random payload lengthens the packet by 128 bits, which on the LE 1M PHY is 128 μ son every one of the two transmissions in every RTT-bearing step. The attack-resistance gain is real; so is the capacity loss, and it is large.

Table 14: Step durations for four representative parameter sets, from Equations (7)–(10). All values in microseconds. Set B is the configuration carried through the worked example of Section 8.

Set	Parameters	$T_{step,0}$	$T_{step,1}$	$T_{step,2}$	$T_{step,3}$	T_{FCS}
A	LE 1M, AA-only RTT, $N_{AP} = 1$, $T_{SW} = 2$, $T_{PM} = 10$, $T_{IP1} = T_{IP2} = 10$	198	108	68	156	15
B	LE 2M, 32-bit sounding, $N_{AP} = 4$, $T_{SW} = 2$, $T_{PM} = 10$, $T_{IP1} = T_{IP2} = 40$	192	134	170	254	80
C	LE 2M, 96-bit sounding, $N_{AP} = 4$, $T_{SW} = 10$, $T_{PM} = 20$, $T_{IP1} = T_{IP2} = 145$	297	303	455	603	150
D	LE 1M, 128-bit random, $N_{AP} = 2$, $T_{SW} = 4$, $T_{PM} = 20$, $T_{IP1} = T_{IP2} = 80$	268	434	234	578	100

Table 14 contains the central design message of this section. The ratio between the cheapest and the most expensive parameter set is a factor of four in Mode-3 and a factor of four in Mode-1, and it is produced entirely by configuration choices that a Host may make for reasons – security margin, antenna coverage, phase averaging – that have nothing to do with scheduling. The capacity consequences are then inherited silently by every subsequent calculation.

7.2 Occupancy of one step

A step occupies its own duration plus the frequency-change period that follows it:

$$T_m^{occ} = T_{step,m} + T_{FCS}. \quad (11)$$

Table 15 gives the resulting overhead fraction. Where T_{FCS} is at the top of its permitted range and the steps are short, more than a third of the subevent is spent retuning.

7.3 Capacity of a subevent

Let the interleaving cadence be n main mode steps per sub mode step, so the mean occupancy of a non-Mode-0 step is

$$\bar{T}^{occ} = \frac{n T_{main}^{occ} + T_{sub}^{occ}}{n + 1}, \quad (12)$$

Table 15: Frequency-change overhead T_{FCS}/T^{occ} for the sets of Table 14.

Set	Mode-0	Mode-1	Mode-2	Mode-3
A	7.0 %	12.2 %	18.1 %	8.8 %
B	29.4 %	37.4 %	32.0 %	24.0 %
C	33.6 %	33.1 %	24.8 %	19.9 %
D	27.2 %	18.7 %	29.9 %	14.7 %

reducing to T_{main}^{occ} when no sub mode is configured. The number of non-Mode-0 steps that fit in a granted subevent is then

$$N_{step}^{(\neq 0)} = \left\lfloor \frac{T_{SE} - N_0 T_0^{occ}}{\bar{T}^{occ}} \right\rfloor, \quad (13)$$

and the total step count of the subevent is $N_{step} = N_0 + N_{step}^{(\neq 0)}$, subject to the specification's own bounds on steps per subevent (representatively 2 to 160) and steps per procedure (representatively 256).

Equation (13) is worth reading twice, because both of its terms are frequently misestimated. The numerator is the granted subevent length *minus the whole Mode-0 block*, which for $N_0 = 3$ in set C is $1341 \mu s$ — more than a millisecond consumed before any distance information is acquired. The denominator is a mean, not the main mode duration: a Mode-2 main mode with a Mode-1 sub mode at $n = 4$ has $\bar{T}^{occ} = (4 \times 250 + 214)/5 = 242.8 \mu s$, so the sub mode makes the average step slightly cheaper, while a Mode-2 main mode with a Mode-3 sub mode makes it dearer.

7.4 Capacity table

Table 16: Non-Mode-0 steps per subevent from Equation (13), for a Mode-2 main mode with a Mode-1 sub mode at cadence $n = 4$, and for a Mode-2 main mode alone. $N_0 = 2$ throughout.

Set	Mode-2 main, Mode-1 sub, $n = 4$			Mode-2 main only		
	$T_{SE} = 1250 \mu s$	$2500 \mu s$	$5000 \mu s$	$1250 \mu s$	$2500 \mu s$	$5000 \mu s$
A	9	22	50	9	24	55
B	2	8	18	2	7	17
C	0	2	7	0	2	6
D	1	4	11	1	5	12

The first row of Table 16 shows what a permissive capability intersection buys: set A fits 50 measurement steps into a 5 ms subevent, enough to cover a substantial fraction of a 72-channel map coherently in a single block. Set C fits seven. A ranging system whose accuracy specification assumes a wide frequency aperture is therefore, implicitly, a system with a requirement on the peer's declared timing capabilities – and that requirement is discovered at capability exchange, long after the accuracy specification was written.

The minimum subevent length is 1250 μs , and it is often not enough. The specification's lower bound on `Min_Subevent_Len` is a legal value, not a useful one. For parameter set C it buys a subevent that cannot hold even the Mode-0 block and two measurement steps, so the enable fails rather than degrades. Hosts that pass the minimum through as their request – because it is the smallest legal value and therefore looks safe – produce campaigns in which the Mode-0 overhead dominates, every procedure is fragmented across many subevents, and demanding capability intersections are rejected outright.

8 Worked scheduling example

This section carries one configuration from Host request to measured behaviour. Every number is derived from the equations of the preceding sections, and Appendix B restates the intermediate values as a check list.

8.1 The scenario

A Central acting as CS initiator maintains a 50 ms connection to a Peripheral acting as reflector. The application wants a distance estimate several times per second, at the best accuracy the pair can support, with a security posture strong enough to resist a distance-reduction attempt. Both devices support two antenna elements, the LE 2M PHY for CS_SYNC, and the fast end of the timing parameter sets.

8.2 Inputs

8.3 Step durations

From Equations (7)–(10) with $T_{SY}^{(0)} = 26 \mu s$ on the LE 2M PHY and $T_{SY} = 42 \mu s$ for the 32-bit sounding sequence:

$$\begin{aligned} T_{step,0} &= 2(26) + 2(5) + 40 + 10 + 80 = 192 \mu s, & T_0^{occ} &= 272 \mu s, \\ T_{step,1} &= 2(42) + 2(5) + 40 = 134 \mu s, & T_1^{occ} &= 214 \mu s, \\ T_{step,2} &= 2(2 + 10)(4 + 1) + 2(5) + 40 = 170 \mu s, & T_2^{occ} &= 250 \mu s. \end{aligned}$$

The mean occupancy of a non-Mode-0 step at $n = 4$, from Equation (12), is $\bar{T}^{occ} = (4 \times 250 + 214)/5 = 242.8 \mu s$.

8.4 The schedule the Controller returns

8.5 Subevent content

Equation (13) with $T_{SE} = 5000 \mu s$ and $N_0 = 2$:

$$N_{step}^{(\neq 0)} = \left\lfloor \frac{5000 - 2(272)}{242.8} \right\rfloor = \left\lfloor \frac{4456}{242.8} \right\rfloor = 18.$$

A full subevent therefore contains 20 steps: 2 Mode-0, 15 Mode-2 and 3 Mode-1. The occupancy check is $2(272) + 15(250) + 3(214) = 4936 \mu s$, leaving $64 \mu s$ of the granted $5000 \mu s$ unused – a 98.7% packing efficiency. The Mode-0 overhead from Equation (5) is $\eta_0 = 544/5000 = 10.9\%$.

Of the 18 non-Mode-0 steps, one is the repeated main mode step required by $R_m = 1$ in every subevent after the first. New channels per subevent are therefore 18 in the first subevent and 17 thereafter.

Table 17: Configuration and procedure parameters as requested by the Host.

Parameter	Value	Rationale
ACL_Connection_Interval	$40 \times 1.25 \text{ ms} = 50 \text{ ms}$	Existing link; not changed for CS
Main_Mode_Type	Mode-2 (PBR)	Precision comes from phase
Sub_Mode_Type	Mode-1 (RTT)	Ambiguity resolution and integrity
Min/Max_Main_Mode_Steps	4 / 4	Fixed cadence $n = 4$ for predictability
Mode_0_Steps	2	One redundant frequency estimate per subevent
Main_Mode_Repetition	1	Procedure will span several subevents
RTT_Type	32-bit sounding sequence	Attack resistance at moderate packet cost
CS_SYNC_PHY	LE 2M	Halves T_{SY}
Channel_Map	72 channels enabled	Maximum aperture
Channel_Map_Repetition	1	One traversal per procedure
Tone_Antenna_Config_Selection	7 (2:2) $\Rightarrow N_{AP} = 4$	Four antenna paths
T_{SW}, T_{PM}	$2 \mu\text{s}, 10 \mu\text{s}$	Fast switch, single-length measurement window
$T_{IP1}, T_{IP2}, T_{FCS}$	$40 \mu\text{s}, 40 \mu\text{s}, 80 \mu\text{s}$	Capability intersection
Min/Max_Subevent_Len	$2500 / 5000 \mu\text{s}$	Ask for a long coherent block
Min/Max_Procedure_Interval	4 / 16 connection intervals	200 ms to 800 ms acceptable
Max_Procedure_Len	$200 \times 0.625 \text{ ms} = 125 \text{ ms}$	Generous headroom
Max_Procedure_Count	0 (continuous)	Tracking campaign

8.6 Procedure content

The channel map has 72 enabled channels and one repetition, so the procedure ends when 72 distinct channels have been measured:

$$18 + 17 + 17 + 17 = 69 \quad \text{after four subevents,} \quad 69 + 3 = 72 \quad \text{three steps into the fifth.}$$

The fifth subevent is therefore truncated: it executes its 2 Mode-0 steps, its repeated main mode step and 3 further main mode steps, then the procedure completes. Totals are given in Table 19.

8.7 Reading the result

Four observations follow from Table 19, and each of them is a design lever.

The aperture is complete but not coherent. All 72 channels are measured, but they are spread over five subevents and therefore five frequency references. The four repeated main mode steps – one per subevent boundary – are what permit the five blocks to be stitched rather than merely averaged. Removing R_m would recover four steps per procedure and destroy that capability.

The procedure spans 101.5 ms of wall clock but only 21.3 ms of scheduler time. The remaining 80 ms is the gap between CS events, imposed by the 50 ms connection interval. If

Table 18: Settled schedule reported at procedure enable, and what it implies.

Reported field	Value	Implication
Subevent_Len	5000 μ s	Maximum requested granted; one coherence block of 5 ms
Subevents_Per_Event	2	Two coherence blocks per ACL anchor
Subevent_Interval	$10 \times 625 \mu$ s = 6250 μ s	CS event spans 11.25 ms of the 50 ms interval
Event_Interval	1	A CS event at every connection event
Procedure_Interval	8	Procedure period = $8 \times 50 = 400$ ms
Procedure_Count	0	Continuous until disabled
Max_Procedure_Len	$200 \times 0.625 =$ 125 ms	Granted as requested

Table 19: Derived campaign quantities for the worked example.

Quantity	Value	Origin
Subevents per procedure	5	Channel sequence exhaustion
CS events per procedure	3	$\lceil 5/2 \rceil$ at $N_{SE} = 2$
Steps per procedure	86	$4 \times 20 + 6$
of which Mode-0	10	$5 \times N_0$
of which Mode-2 (PBR)	64	main mode, incl. 4 repeats
of which Mode-1 (RTT)	12	sub mode, 3 per full subevent
Distinct channels measured	72	Full map traversed
Frequency aperture	71 MHz	72 channels at 1 MHz spacing
Procedure wall-clock span	101.5 ms	$2T_C$ + partial third event
Granted procedure length bound	125 ms	Not binding
Radio-active time per procedure	14.41 ms	$\sum T_{step}$ over 86 steps
Scheduler-occupied time per procedure	21.29 ms	$\sum T^{occ}$ over 86 steps
Procedure period	400 ms	$N_{PI}T_C$
Update rate	2.5 Hz	1/400 ms
Radio duty cycle	3.6 %	14.41/400
Scheduler duty cycle	5.3 %	21.29/400
Mean radio current at 6 mA active	0.22 mA	duty cycle $\times$ 6 mA

target motion matters, this is the number that matters: a target moving at 1.5 m/s displaces 15 cm across the aperture, which is larger than the ranging precision the configuration is capable of. Shortening the connection interval – an ACL change, not a CS change – is the only way to compress it.

Twelve RTT steps disambiguate 64 PBR steps. At $n = 4$ the sub mode consumes 20 % of the non-Mode-0 step budget for a measurement whose precision is orders of magnitude coarser. That is the correct trade when ambiguity must be resolved reliably; where a prior range estimate is available from the previous procedure, $n = 9$ would recover roughly six PBR steps per procedure.

The update rate is quantised. $N_{PI} = 8$ gives 400 ms. The neighbouring options are $N_{PI} = 7$ (350 ms) and $N_{PI} = 9$ (450 ms). There is no 375 ms, because the connection interval is the quantum. An application specification written as “2.7 Hz” cannot be met on this link; one written as “at least 2.2 Hz” can.

The one-subevent alternative. Had the Host requested and been granted a single subevent long enough to hold all 72 channels – roughly $72 \times 242.8 + 544 = 18\,026 \mu s$ – the procedure would have had one frequency reference, no repetition overhead, and a 18 ms span instead of 101.5 ms. It would also have required a subevent occupying 36 % of the connection interval, which most Controllers will decline on a link carrying any other traffic. The tension between coherence and connection-interval occupancy is the central scheduling trade in Channel Sounding, and it has no general resolution: it is settled per link, by the Controller, at enable.

9 Channel map, selection and coexistence

9.1 The CS channel set

Channel Sounding uses the 2.4 GHz ISM band on a 1 MHz raster, indexed so that CS channel k sits at $2402 + k$ MHz for $k = 0 \dots 78$. Seven indices are permanently excluded, being those that coincide with the LE primary advertising channels and the band edges: $k \in \{0, 1, 23, 24, 25, 77, 78\}$. Seventy-two channels therefore remain available, and the specification requires that at least fifteen of them be enabled in any configuration’s channel map.

The floor of fifteen is a floor on usability, not on accuracy. The Mode-2 report shows ranging variance from a phase-slope fit falling with both the number of channels and the square of the frequency span; a fifteen-channel map spanning 15 MHz and a seventy-two-channel map spanning 71 MHz are not comparable instruments. Channel map size is therefore an accuracy parameter that lives in the configuration, and is chosen before any measurement is made.

9.2 Two channel maps, three mechanisms

Three distinct channel-related mechanisms operate at once, and conflating them is a common source of confusion.

Table 20: The three channel mechanisms and their scopes.

Mechanism	Scope	Set by	Takes effect
ACL channel map	The connection’s data channels	Central; adaptive frequency hopping	Connection update instant
CS <code>Channel_Map</code>	Which CS channels the configuration may use	Configuration creation	At configuration creation
CS channel classification	Which of those are currently believed bad	Either device, during the campaign	At a procedure boundary

The third row carries the operational subtlety. Classification is a running input: a device that detects a jammer or a persistent interferer reports it, and the campaign avoids those channels. But the update does not take effect immediately – it takes effect at a procedure boundary, because the channel sequence for a procedure is generated once, at the procedure’s start, and consuming it is what defines the procedure. A classification update issued mid-procedure influences the next procedure, not the current one. For a 400 ms procedure period this means up to 400 ms of continued measurement on a channel already known to be bad, and the per-step validity gates – not the channel map – are what protect the estimate in the meantime.

9.3 Channel selection algorithms #3b and #3c

Both algorithms produce a pseudo-random permutation of the enabled channels, seeded from the campaign’s cryptographic material so that the sequence is unpredictable to an observer. They differ in what they optimise.

CSA #3b produces a shuffled traversal in which every enabled channel is visited once before any is visited again. Its property is uniform coverage: after one traversal the aperture is complete and unbiased, which is exactly what a phase-slope fit wants.

CSA #3c produces a shaped traversal, controlled by `Ch3c_Shape` (a “hat” or an “X” pattern) and `Ch3c_Jump` (the step size between successive selections, 2 to 8). Its property is a controlled ordering in frequency: consecutive steps are separated by a known jump rather than landing arbitrarily. This matters for two reasons. First, a large frequency jump between consecutive steps stresses the synthesiser more than a small one, and T_{FCS} is a fixed budget: a shaped sequence keeps the retune within a settled range. Second, a procedure that is truncated before it completes leaves a partial aperture, and a shaped sequence makes that partial aperture predictable rather than arbitrary.

Truncation interacts with channel selection. Section 5 listed three ways a procedure can end early. Under *CSA #3b* a truncated procedure has measured a random subset of the map, and the effective span of that subset varies from fix to fix – which makes the ranging precision vary from fix to fix in a way the aggregation layer cannot see unless it computes the realised span. Under *CSA #3c* the subset is structured. Systems that expect frequent truncation should prefer the shaped algorithm and should compute per-fix precision from the channels actually reported, never from the configured map.

9.4 Coexistence with the connection and with other traffic

The campaign is not the only user of the radio. Three coexistence pressures act on it.

The connection itself. The connection events must occur; CS activity is placed around them. This is the mechanism behind a Controller granting $N_{EI} = 2$ where $N_{EI} = 1$ was acceptable to the Host, and behind a granted subevent shorter than the requested one.

Other links. A Controller maintaining several connections, an advertising set, or a periodic sync must interleave them all. A CS event is a comparatively long, non-interruptible commitment, and a Controller may legitimately decline a schedule that a single-link device would accept. Campaign parameters that work in bench testing against one peer are therefore not guaranteed to work in a deployment where the same device holds four connections.

Non-Bluetooth users of the band. Wi-Fi occupancy is the dominant external pressure, and it is not uniform across the CS channel set: the three common 20 MHz Wi-Fi channels occupy large contiguous blocks. A classification input derived from the connection’s own adaptive frequency hopping statistics is a good first approximation and costs nothing, since the connection is already measuring it.

10 Results, completion status and abort

10.1 What is reported and when

Results are delivered per subevent, not per procedure and not per step. Each subevent produces a result event carrying the campaign context and a list of per-step results; where the list is too large for one event it is continued in follow-on events. The Host therefore observes the campaign at the coherence-block granularity, which is the correct granularity: it receives a set of steps that share a frequency reference, together with the metadata needed

to know whether they are usable.

Table 21: Campaign-level fields carried with each subevent result, and their use.

Field	Use in the aggregation layer
Connection_Handle, Config_ID	Binds the result to a peer and a configuration
Start_ACL_Conn_ Event_Counter	Places the subevent on the shared timeline; the basis of any timestamping of the fix
Procedure_Counter	Groups subevents into procedures; the join key for a per-fix aggregation
Frequency_Compensation	The subevent's frequency reference, from the Mode-0 steps
Reference_Power_Level	Normalisation for the per-step power and quality metrics
Procedure_Done_Status	Whether the procedure is complete, partial, or aborted
Subevent_Done_Status	The same for this subevent
Abort_Reason	Why, if either is aborted
Num_Antenna_Paths	Number of paths per Mode-2/3 step in this result
Num_Steps_Reported	Length of the per-step list that follows

Per step, the result carries the step mode, the channel index, the step's data length, and the mode-specific payload analysed in the companion reports: the received frequency and packet quality for Mode-0, the timestamps or the RTT with its quality and attack-detector metric for Mode-1, the per-path phase correction terms with their quality indicators for Mode-2, and both for Mode-3.

10.2 Completion status and abort

The completion status is the field that distinguishes a shorter result from a wrong one, and an aggregation layer that ignores it will silently mix the two.

Table 22: Completion outcomes and the correct response.

Outcome	Meaning	Correct handling
Complete	The subevent or procedure ran as scheduled	Aggregate normally
Partial, continuing	More results follow for this subevent or procedure	Buffer; do not fix yet
Aborted: local scheduling	The Controller could not run the activity	Aggregate what arrived, with the realised aperture; do not treat as a link fault
Aborted: no sync	Mode-0 synchronisation failed for the subevent	Discard the whole subevent: its steps have no valid frequency reference
Aborted: channel map update	Classification changed	Expect a re-generated sequence next procedure
Aborted: peer request	The peer terminated the campaign	Stop; re-enable requires a new negotiation

The second abort row is the one with a hard consequence. A subevent whose Mode-0 steps failed contains steps that were nonetheless executed and reported, and those steps carry phase and timing values that look ordinary. They are not usable: the frequency reference on

which the Mode-1 clock correction and the Mode-2 drift term depend was never established. Discarding them is not conservative practice, it is correctness.

11 Campaign-level error and robustness

The mode reports develop per-step error budgets. This section covers the error terms that exist only because the campaign is a structure in time.

Table 23: Campaign-level error and robustness terms. Magnitudes use the worked example of Section 8 unless stated.

Term	Mechanism	Magnitude and control
Aperture smearing	Target moves during the procedure; channels measured at different positions	15 cm at 1.5 m/s over 101.5 ms; control by shortening T_C or the procedure
Inter-subevent reference step	Each subevent has its own Mode-0 reference; combining across them adds their difference	Bounded by Mode-0 precision; control with $R_m \geq 1$ and per-subevent fits
Result staleness	Age of the newest fix when the application uses it	Up to one procedure period plus report latency: 400 ms here
Silent staleness on link loss	Campaign ends with the connection; last fix keeps its apparent validity	Bounded by the supervision timeout, not by the procedure period
Aperture variability	Truncated procedures measure fewer channels; precision varies per fix	Compute the realised span per fix; prefer CSA #3c where truncation is common
Classification lag	A bad channel keeps being measured until the next procedure	Up to one procedure period; per-step validity gates absorb it
Cadence variability	Controller may vary n within $[n_{\min}, n_{\max}]$	Count reported step modes rather than predicting them
FAE table staleness	Cached peer calibration ages with temperature	Refetch on long campaigns or on a temperature excursion
Schedule drift after ACL update	Intervals are in connection intervals; T_C changed	Re-read the settled schedule after any connection parameter update
Security context change	Re-pairing invalidates the CS nonce material	Re-run security establishment; never cache across it

11.1 Choosing the procedure period

Two competing quantities set the procedure period. Staleness argues for a short period; energy, coexistence and estimator independence argue for a long one. The third is easy to overlook: two procedures separated by less than the channel coherence time of the environ-

ment produce correlated errors, so a doubled update rate does not halve the variance of a smoothed track. Where the campaign feeds a tracking filter, the filter's process noise should be set from the true procedure period and the innovation sequence should be checked for the correlation that too short a period produces.

11.2 Degrading gracefully

A campaign that cannot be scheduled as requested should degrade along a chosen axis rather than whichever axis the Controller happens to relax. The order that preserves the most useful behaviour is usually: first lengthen the procedure interval, accepting a lower update rate; then reduce the sub mode cadence, accepting weaker disambiguation; then reduce the channel map, accepting lower precision; and only last reduce the subevent length, because that fragments the coherence structure and its effect on the result is the hardest to model.

12 Verification and conformance-oriented test plan

12.1 Layered verification

Campaign behaviour is verified in four layers, each of which can be exercised without the one below it being correct.

Layer 1 – parameter conformance, no radio. Drive the configuration lifecycle against a Controller with the radio disabled or in a loopback mode, and check that every parameter is accepted, rejected or coerced as the specification requires. The cases that matter are the boundary ones: an illegal main/sub mode pair; `Mode_0_Steps` outside 1–3; a channel map with fewer than fifteen enabled channels; a channel map that enables an excluded index; $n_{\min} > n_{\max}$; `Min_Subevent_Len` greater than `Max_Subevent_Len`; a procedure interval range that excludes every achievable value. A Controller that silently accepts any of these and then behaves unexpectedly on air is a much harder fault to find later.

Layer 2 – schedule prediction. With the campaign enabled, capture the settled schedule and predict, from Section 7, the step count per subevent, the subevent count per procedure and the procedure span. Compare with the reported results. A mismatch here isolates cleanly: a wrong step count means a step-duration or T_{FCS} assumption is wrong; a wrong subevent count means a channel-sequence assumption is wrong; a wrong span means the event structure is wrong.

Layer 3 – air-interface timing. With a spectrum or protocol analyser, confirm that CS events occur at the predicted offset from the connection anchor points, that subevents are spaced by the reported interval, that T_{FCS} separates consecutive steps, and that the step ordering follows Section 6 – Mode-0 first, then repetition, then the interleaved pattern. This is the layer at which a mis-implemented `Main_Mode_Repetition` becomes visible, because the repeated steps re-use channels that the sequence has already produced.

Layer 4 – end-to-end at a known distance. Run the campaign against a peer at a surveyed distance, in a low-multipath environment, and check the ranging output. This layer validates the combination of everything; it is the least diagnostic and should be run last.

12.2 Test cases specific to the campaign

12.3 Minimum acceptance records

For each accepted configuration, the following should be recorded and retained, because together they make a ranging result reproducible: the full configuration and procedure pa-

Table 24: Campaign test cases that per-step verification does not reach.

Case	Stimulus	Expected observation
Connection parameter update mid-campaign	Change T_C while a continuous campaign runs	Campaign is re-negotiated, terminated, or the update is refused – never silently re-timed
Peripheral latency interaction	Enable latency, then a campaign	CS events still occur at their anchors; the Peripheral wakes for them
Subevent grant below request	Request <code>Min.Subevent.Len</code> that the link cannot accommodate	Enable fails, or a shorter subevent is granted and reported; never a silent overrun
Classification during a procedure	Disable a channel mid-procedure	Current procedure unaffected; next procedure’s sequence excludes it
Mode-0 failure	Attenuate the peer during the Mode-0 steps of one subevent	That subevent reports an abort; its steps are not presented as valid
Procedure truncation	Set <code>Max.Procedure.Len</code> below the procedure’s natural length	Procedure completes early with a partial status and a reduced channel set
Re-pairing mid-campaign	Force a security context change	Campaign stops; re-enable requires fresh security establishment
Continuous campaign longevity	Run continuously for hours	Procedure counter wraps handled; no drift in the realised procedure period
Capability intersection	Peer declaring only slow timing values	Step count per subevent falls as Table 16 predicts

parameter sets as requested; the settled schedule as reported; the capability sets of both devices; the channel map and classification history; and, per procedure, the realised step count, mode histogram, channel list and completion status. Ranging results archived without the realised channel list cannot have their precision recomputed, and are therefore not auditable.

13 Normative behaviour versus implementation choice

Campaign behaviour mixes strictly specified structure with wide Controller discretion, and knowing which is which prevents both over-constrained designs and misplaced conformance expectations.

The practical reading of this table is that a Host may specify *what* is measured with confidence, and must *observe* when it is measured. The configuration parameters — modes, RTT type, channel map, cadence bounds, Mode-0 count, repetition — are agreed and honoured. The scheduling parameters are requests. Portable ranging software is written accordingly: it fixes the configuration, reads the schedule, and derives every timing and precision expectation from what was reported rather than from what was asked.

Table 25: The boundary between specified behaviour and implementation choice at campaign level.

Specified	Left to the implementation
Step ordering within a subevent: Mode-0 first, then repetition, then interleaved main and sub mode	The cadence actually chosen within $[n_{\min}, n_{\max}]$, and whether it is constant
The permitted value sets for T_{IP1} , T_{IP2} , T_{SW} , T_{PM} , T_{FCS} and the fixed T_{RD} , T_{GD} , T_{FM}	Which of the permitted values a device declares as supported
Units and ranges of every scheduling parameter	Which value within the Host's requested range the Controller grants
That a procedure is anchored at an offset from an ACL connection event anchor point	The offset chosen, and the negotiation strategy used to agree it
That the channel sequence is generated by the selected algorithm from the campaign's random material	Scheduling of the sequence across subevents and events
That classification takes effect at a procedure boundary	How a device forms its classification in the first place
That results are reported per subevent with completion status	Buffering, batching and continuation policy for large result sets
That a lost connection terminates the campaign	Whether and how the Host is helped to restart it
Mandatory support for Modes 0, 1 and 2; optional Mode-3	Whether Mode-3, CSA #3c, the companion signal or SNR control are supported at all

14 Conclusions

A Channel Sounding campaign is a negotiated structure, not a command. The Host states what it wants measured and within what bounds it can be scheduled; the Controllers, over the ACL connection, agree a configuration and settle a schedule; and only the settled schedule describes what happens on air.

The configuration lifecycle has eight stages, four of which are Link Layer exchanges with the peer and are therefore subject to the connection's own latency and reliability. The capability intersection reached in the second stage silently determines the step durations that govern everything afterwards. Security establishment is not optional infrastructure: it seeds the access addresses, sequences, channel order and antenna permutation that make a campaign resistant to distance reduction, and it is invalidated by any change to the connection's security context.

The ACL connection enters the campaign in four capacities: as the time base whose anchor points every CS event hangs from, as the quantum in which event and procedure intervals are counted — and therefore as the quantiser of the achievable update rate, as the root of the campaign's cryptographic material, and as the liveness relationship whose loss ends the campaign without a completion event. Every one of the mappings in Table 10 is a mechanism, not a correlation, and an application that changes connection parameters without re-reading its schedule has changed its own measurement rate.

The scheduling hierarchy has five levels, and the subevent is the one that matters most to estimation: it is the block within which a single Mode-0 frequency reference is valid, and therefore the block within which phase measurements combine cleanly. Inside it, the ordering is fixed — one to three Mode-0 steps, then the steps repeated from the previous subevent for

continuity, then main mode steps with sub mode steps interleaved at a controlled cadence — while the exact cadence and the granted length are the Controller’s. The per-mode step durations of Equations (7)–(10), together with the frequency-change period, convert that granted length into a step count, and the step count into a frequency aperture.

The worked example shows the whole chain in numbers: a 50 ms connection, a Mode-2 main mode with a Mode-1 sub mode, and a 5 ms subevent yield 18 measurement steps per coherence block, five blocks per procedure, 72 channels of aperture, an update rate of 2.5 Hz and a radio duty cycle of 3.6 %. It also shows the trade that cannot be evaded: the same procedure spans 101.5 ms of wall clock, of which only 21.3 ms is scheduler time, because the connection interval separates the CS events. Coherence, aperture, update rate and connection occupancy pull against each other, and the campaign is where that tension is resolved.

A Host interface reference

This appendix tabulates the Host interface of the campaign as a single reference. Command and event names follow the cited revision; parameter details are those used in the body of this report.

A.1 Commands, in lifecycle order

Table 26: LE CS commands. “LL” marks commands that generate a Link Layer exchange with the peer and therefore complete asynchronously, after a completion event.

Command	LL	Purpose and principal parameters
LE CS Read Local Supported Capabilities	–	Returns Table 4 for this Controller
LE CS Read Remote Supported Capabilities	yes	Fetches and caches the peer’s capabilities
LE CS Write Cached Remote Supported Capabilities	–	Installs a previously obtained capability set without an air exchange
LE CS Set Default Settings	–	<code>Role_Enable</code> , <code>CS_SYNC_Antenna_Selection</code> , <code>Max_TX_Power</code>
LE CS Security Enable	yes	Establishes <code>CS_IV</code> , <code>CS_IN</code> , <code>CS_PV</code> from the connection’s security context
LE CS Read Remote FAE Table	yes	Fetches the peer’s per-channel frequency actuation error table
LE CS Write Cached Remote FAE Table	–	Installs a cached FAE table
LE CS Create Config	optional	Table 5; <code>Create_Context</code> decides whether the peer is configured too
LE CS Remove Config	–	Releases a <code>Config_ID</code>
LE CS Set Channel Classification	yes	10-octet classification bitmap
LE CS Set Procedure Parameters	–	Table 7
LE CS Procedure Enable	yes	<code>Connection_Handle</code> , <code>Config_ID</code> , enable or disable
LE CS Test / LE CS Test End	–	Single-device test mode; outside the scope of this report

A.2 Events

Table 27: LE CS events and what the Host should do with each.

Event	Content and use
LE CS Read Remote Supported Capabilities Complete	The peer's capability set; compute the intersection and fix the timing parameters
LE CS Read Remote FAE Table Complete	The peer's FAE table; cache with a validity policy
LE CS Security Enable Complete	Security material established; configurations using sounding or random RTT types may now be enabled
LE CS Config Complete	The configuration as agreed, echoed back; compare against what was requested
LE CS Procedure Enable Complete	Table 8: the settled schedule. Derive all timing expectations here
LE CS Subevent Result	Table 21 plus the per-step list
LE CS Subevent Result Continue	Continuation of an oversized result list; same procedure and subevent context

A.3 Units, collected

Table 28: Every campaign quantity with its unit. Unit confusion between 0.625 ms, 1.25 ms, 625 μ s and connection intervals is the most common source of scheduling arithmetic errors.

Quantity	Unit	Multiply by
ACL connection interval	1.25 ms	1.25 ms
Max_Procedure_Len	0.625 ms	0.625 ms
Min/Max_Procedure_Interval	connection intervals	T_C
Event_Interval, Procedure_Interval	connection intervals	T_C
Subevent_Interval	625 μ s	625 μ s
Min/Max_Subevent_Len, Subevent_Len	microseconds	1
Step timing parameters	microseconds	1
Procedure_Count, Subevents_Per_Event	counts	1

B Worked-example values for regression use

The values below reproduce Section 8 in a form suitable for a unit test of a scheduling calculator. All durations are microseconds unless stated.

Table 29: Inputs.

Quantity	Value	Note
T_C	50 000	40×1.25 ms
$T_{SY}^{(0)}$	26	LE 2M, no sequence
T_{SY}	42	LE 2M, 32-bit sounding sequence
T_{RD}, T_{GD}, T_{FM}	5, 10, 80	Fixed
$T_{IP1}, T_{IP2}, T_{FCS}$	40, 40, 80	Negotiated
T_{SW}, T_{PM}, N_{AP}	2, 10, 4	ACI 7
N_0, R_m, n	2, 1, 4	Configuration
T_{SE}, T_{SEI}, N_{SE}	5000, 6250, 2	Granted
N_{EI}, N_{PI}	1, 8	Granted, in connection intervals
Enabled channels, repetition	72, 1	Channel map

Table 30: Expected outputs. A calculator that reproduces this column is consistent with Sections 6 to 8.

Quantity	Value	Equation
$T_{step,0}, T_0^{occ}$	192, 272	(7), (11)
$T_{step,1}, T_1^{occ}$	134, 214	(8), (11)
$T_{step,2}, T_2^{occ}$	170, 250	(9), (11)
$T_{step,3}$ (if Mode-3 were used)	254	(10)
$\bar{T}^{occ}$	242.8	(12)
Non-Mode-0 steps per full subevent	18	(13)
Steps per full subevent	20	$N_0 + 18$
Mode composition of a full subevent	2 / 15 / 3	Mode-0 / Mode-2 / Mode-1
Subevent occupancy	4936	$2(272) + 15(250) + 3(214)$
Packing efficiency	98.7 %	$4936/5000$
Mode-0 overhead η_0	10.9 %	(5)
New channels: first / later subevents	18 / 17	R_m inert in the first
Subevents per procedure	5	$18 + 17 + 17 + 17 + 3 = 72$
CS events per procedure	3	$\lceil 5/2 \rceil$
Steps per procedure	86	$4(20) + 6$
Mode histogram per procedure	10 / 64 / 12	Mode-0 / Mode-2 / Mode-1
Radio-active time per procedure	14 408	$\sum T_{step}$
Scheduler-occupied time per procedure	21 288	$\sum T^{occ}$
Procedure wall-clock span	101 544	$2T_C + 1544$
Procedure period	400 000	$N_{PI}T_C$
Radio duty cycle	3.60 %	$14\,408/400\,000$
Scheduler duty cycle	5.32 %	$21\,288/400\,000$
Single-subevent alternative length	18 026	$72\bar{T}^{occ} + N_0T_0^{occ}$

References

- [1] Bluetooth SIG, *Bluetooth Core Specification, Version 6.3*, Vol. 1, Part A, “Architecture, Change History, and Conventions,” Sec. 9.1 “Channel Sounding procedure,” adopted 2026.
https://www.bluetooth.com/wp-content/uploads/Files/Specification/HTML/Core_v6.3/out/en/architecture,-change-history,-and-conventions/architecture.html
- [2] Bluetooth SIG, *Bluetooth Core Specification, Version 6.3*, Vol. 6, Part H, “Channel Sounding,” Secs. 4.1–4.2 (procedure, event, subevent and step scheduling), Sec. 4.3 (step modes and step timing) and Sec. 4.5 (antenna configuration and channel selection), adopted 2026.
https://www.bluetooth.com/wp-content/uploads/Files/Specification/HTML/Core_v6.3/out/en/low-energy-controller/channel-sounding.html
- [3] Bluetooth SIG, *Bluetooth Core Specification, Version 6.3*, Vol. 6, Part B, “Link Layer Specification,” CS capability, security, configuration and procedure control procedures, adopted 2026.
- [4] Bluetooth SIG, *Bluetooth Core Specification, Version 6.3*, Vol. 4, Part E, “Host Controller Interface Functional Specification,” LE CS command and event set, adopted 2026.
- [5] Bluetooth SIG, “Bluetooth Core 6.3 Technical Overview,” 2026.
<https://www.bluetooth.com/bluetooth-core-6-3-technical-overview/>
- [6] Bluetooth SIG, “Bluetooth Channel Sounding” feature page and Core 6.0 feature overview.
<https://www.bluetooth.com/learn-about-bluetooth/feature-enhancements/channel-sounding/>
- [7] Silicon Labs, “Bluetooth LE Channel Sounding Fundamentals,” Location Services documentation (CS procedure, event, subevent and step hierarchy; CS offset from the ACL anchor point).
<https://docs.silabs.com/rtl-lib/latest/rtl-lib-channel-sounding-fundamentals/>
- [8] Silicon Labs, “Bluetooth Stack API Reference: CS commands,” parameter ranges and units for `sl_bt_cs_set_procedure_parameters` and `sl_bt_cs_create_config`.
<https://docs.silabs.com/bluetooth/latest/bluetooth-stack-api/sl-bt-cs>
- [9] Texas Instruments, “Channel Sounding,” SimpleLink Low Power F3 SDK BLE5-Stack User’s Guide (configuration command sequence).
https://software-dl.ti.com/simplelink/esd/simplelink_lowpower_f3_sdk/9.12.00.19/exports/docs/ble5stack/ble_user_guide/html/channel-sounding/channel-sounding.html
- [10] MathWorks, “What Is Bluetooth Channel Sounding?,” Bluetooth Toolbox documentation (procedure, event, subevent and step bounds).
<https://www.mathworks.com/help/bluetooth/ug/what-is-bluetooth-channel-sounding.html>
- [11] Zephyr Project, “Channel Sounding (CS)” API documentation, `bt_le_cs_*` structures and the procedure enable completion fields.
https://docs.zephyrproject.org/latest/doxygen/html/group__bt__le__cs.html
- [12] Texas Instruments, “High Accuracy, Low Cost, Secure Ranging with Bluetooth,” application report SWRA791.

<https://www.ti.com/lit/swra791>

- [13] *Bluetooth CS Mode-0 Frequency Estimation*, Revision 3.0. Companion report: Mode-0 step structure, T_{GD} and T_{FM} , subevent frequency reference.
- [14] *Bluetooth CS Mode-1 RTT Estimation*, Revision 1.0. Companion report: Mode-1 step structure, T_{SY} by RTT type and PHY, sounding sequence and attack detector.
- [15] *Bluetooth CS Mode-2 Phase-Based Ranging*, Revision 1.0. Companion report: Mode-2 step structure, tone slots, antenna configuration index, aperture and ranging precision.